Measuring the Milky Way's Rotation Curve from HI4PI 21 cm Data

Project report — tangent-point method, Quadrant I *Name: Roshani · Date: 6 September 2026 · Data run executed end-to-end on this machine (see `run_record.json`)*

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1. Introduction

A **rotation curve** is the orbital speed of gas around the Galactic Centre as a function of distance from it. It is the single most direct way to weigh a galaxy: at every radius, the speed tells us how much gravitating mass lies inside.

Measuring it from inside the Galaxy is hard. We cannot step outside and watch; we sit roughly 8 kpc from the centre, and every measurement must be recovered from the motion of gas **along our own lines of sight**. The experiment therefore relies on the Galaxy rotating *differentially* — inner gas orbits faster than outer gas — so that gas at different distances along one sightline produces emission at different radial velocities.

The tracer is **neutral hydrogen (H I)**. The electron in a hydrogen atom can flip its spin relative to the proton, emitting a photon at a rest frequency of **1420.405752 MHz** — the 21 cm line, known to better than one part in 10^{11}. H I is abundant across the whole disk, and radio waves pass through the dust that blocks visible light, so the entire disk is observable.

This report measures the rotation curve over Galactocentric radii **R = 2.1–8.0 kpc** using the **tangent-point method** on the HI4PI survey, converts the speeds into an enclosed-mass scale, and tests the result against a Keplerian expectation.

2. Data

Survey. HI4PI — the full-sky 21 cm H I survey (HI4PI Collaboration, Ben Bekhti et al. 2016, A&A 594, A116), merging EBHIS (Effelsberg 100 m, Germany) with GASS (Parkes 64 m, Australia). Quoted properties: 16.2 arcmin resolution, 1.29 km/s channel width, ~43 mK channel noise.

Files used. Five CAR_E cubes covering l = 15°–80°: `CAR_E01.fits` … `CAR_E05.fits` from

https://cdsarc.cds.unistra.fr/ftp/J/A+A/594/A116/CUBES/GAL/CAR (264.1 MB each, 933 × 266 × 266 voxels). Every file's SHA-256 checksum was computed and matches the values recorded in `run_record.json`, so these are exactly the published data.

Frames and units. All velocities are in the **LSR frame** (Local Standard of Rest), in km/s. Brightness is brightness temperature T_B in kelvin. Distances are in kpc.

3. Method

The analysis follows the notebook's fifteen steps; the decision-level summary is:

Setting	Value	Reason
Longitude range	l = 15°–80°	Below 15° the Galactic bar breaks circular rotation; above 80° the terminal-velocity signal becomes too small
Latitude strip	|b| ≤ 0.5°	Confined to the thin H I layer; averaging over latitude raises the signal-to-noise ratio
Longitude bin	0.25°	Median of ~3 sightlines per bin; fine enough to follow the curve, wide enough to average lumps
Velocity window	−150 to +300 km/s	Quadrant I signal lies at positive v_LSR; the window must contain the search range
Edge search window	15–260 km/s	Skips bright local gas near v = 0; nothing inside R_0 orbits faster than ~260 km/s
Threshold	fixed **10 K**	Where the emission "ends"; a decision, not a measurement (see §5)
3-channel rule	≥ 3 consecutive channels above threshold	Rejects single-channel noise spikes at the edge
Interpolation	linear between channels	The channel grid is 1.29 km/s; the exact 10 K crossing is interpolated, not snapped to a channel
Edge-bias calibration	ON, using l ≥ 70°	The one region where geometry fixes the answer, so our own bias can be measured

Setting	Value	Reason
		(§4)

Physics chain. Each sightline's spectrum yields a **terminal velocity** v_term — the extreme-velocity edge of the emission, not the brightest peak, because the edge is set by the fastest orbital motion on the sightline (the dynamical information), while the peak merely traces gas density. Tangent-point geometry then gives, for each sightline:

4. Results

The run measured **780 sightlines; all 780 gave a clean edge (100% flagged OK)**, with v_term ranging 19.5–153.2 km/s, binned into **260 longitude bins**. The quality-control overlay (Fig. 3) shows the measured points hugging the edge of the emission along its full length.

The high-longitude diagnostic (the honest part). At l = 90° the terminal velocity must equal zero, because $\Theta_0 \sin 90°$ exactly cancels $\Theta(R_0)$. Between l = 70° and 90° our measured edges sit systematically high: the **median excess is 13.6 km/s** (uncorrected median Θ = 247.5 km/s; corrected median Θ = 233.8 km/s — both numbers quoted, as required). At the outermost point the uncorrected curve reads 256 km/s, the corrected curve 243 km/s, against the value 236 km/s it must approach by construction. A constant correction shifts the velocity zero point; the *shape* of the curve never changes (Fig. 11).

The rotation curve (Fig. 5): 260 points from **R = 2.13 to 8.02 kpc**, circular speed 196–246 km/s, median 234 km/s. The curve rises from ~200 km/s at 2.1 kpc to a roughly flat ~230–245 km/s beyond 4 kpc, arriving at 242.5 km/s near R_0 — close to the adopted Θ_0 = 236 km/s, as geometry demands.

Mass scale (Fig. 6): $M(<R) = \Theta^2 R/G$ grows from ~2.0×10^{10} $M_\odot$ at 2.1 kpc to **1.1×10^{11} $M_\odot$ at 8 kpc**. Between R = 2.1 and 8.0 kpc the radius grew by a factor 3.8 while the enclosed mass grew by a factor **5.5** — the signature of a flat rotation curve: mass is still being added far out.

Keplerian comparison (Fig. 7): anchored at the innermost point (201 km/s at R = 2.13 kpc), a point-mass curve would fall to **103.5 km/s by R = 8.02 kpc**. We measure **242.5 km/s**. The implied enclosed mass is a factor **5.5** larger than the point-mass model allows. The Galaxy has not run out of mass.

External comparison (Fig. 8): joined to the Eilers et al. (2019) fit for the outer Galaxy, our corrected curve reads 231.8–236.5 km/s over 5–7.8 kpc against Eilers' 229.5–234.3 km/s — agreement at the ~1–2% level, including at 5–6 kpc where

the correction was *not* fitted, which makes that agreement a genuine test.

- $R = R_0 \sin l$ — the tangent point is the sightline's closest approach to the Galactic Centre; trigonometry fixes its radius from longitude alone
- $\Theta(R) = |v_{term}| + \Theta_0 \sin l$ — the Sun's own circular motion is added back; absolute values appear because the sign convention depends on the quadrant

Finally each speed is converted to a spherical-equivalent enclosed mass, $M(<R) = \Theta^2 R/G$, with $G = 4.30091 \times 10^{-6}$ kpc (km/s)² / M☉ so the result emerges directly in solar masses (unit check: a 220 km/s curve at 8 kpc gives 9.0×10^{10} M☉).

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5. Uncertainties — statistical and systematic, kept separate

Statistical (random). The scatter bars on the curve are a fixed **8 km/s**, adopted to represent the scale of turbulent and non-circular gas motions. This is an estimate of physical scatter, not a formal error bar. Each bin is the median of ~3 sightlines.

Systematic — measured by us. The fixed 10 K threshold always lands on the turbulence-smeared falling edge of the emission (the edge is smeared over 8–10 km/s), so it fires systematically *late* and biases v_term *outwards*. The l → 90° limit exposes this: there v_term must be zero, but we measure a median excess of **13.6 km/s**. We subtract this from every sightline and propagate the Θ_0 systematic below.

Systematic — adopted. The constants $R_0 = 8.15 \pm 0.15$ kpc and $\Theta_0 = 236 \pm 7$ km/s (adoption from the notebook Step 2; these are the recommended IAU-era values) set the radius scale and velocity zero point of every point on the curve; $\Theta_0 \pm 7$ km/s appears as the grey systematic band in Fig. 5. These are *not* measured by this experiment and must always be quoted with the result.

The analyst's share. Re-running the whole extraction under seven different edge criteria (`criterion_sensitivity.csv`) gives medians spread over **15 km/s before correction and 5 km/s after** — while the telescope's own channel noise is 0.043 K, two orders of magnitude below the 10 K threshold. Almost none of the spread comes from the instrument; it comes from decisions the analyst makes. Keeping statistical and systematic separate is what makes that visible.

6. Conclusion

From the HI4PI 21 cm survey and the tangent-point method we measured the Milky Way's rotation curve over R = 2.13–8.02 kpc. The curve rises from ~200 km/s to a roughly flat 230–245 km/s, and the enclosed mass grows from 2×10^{10} to 1.1×10^{11} $M_\odot$ — a factor 5.5 above what a Keplerian point-mass model anchored at the innermost radius allows.

The conclusion this measurement supports is that the **gravitating mass distribution of the Milky Way extends far beyond the region where most of the starlight is**; the spherical-equivalent mass keeps increasing with radius out to at least the Sun's orbit, and the comparison with published outer-Galaxy

work shows rotation staying far from Keplerian out to 25 kpc. This is evidence for an **extended gravitating mass distribution**, not a dark-matter detection: the tangent-point method stops at R_0, and a dark-matter claim would additionally require a baryonic mass model (stars, gas, dust) and outer-Galaxy tracers showing the residual mass cannot be baryonic.

Reproduction notes

- Full run executed 6 September 2026; notebook `MilkyWay_Rotation_Curve_ALGO.ipynb`, executed copy `executed_full_run.ipynb`; all settings, constants, checksums and software versions in `mw_rotation_results/run_record.json`.
- First (fast) validation run with LON_MAX = 40: 300 sightlines, R = 2.13–5.23 kpc, median Θ = 228 km/s, all flags OK. In that range the edge-bias calibration correctly reports 0 — the calibration region (l ≥ 70°) lies outside the observed range, so the full 15–80° run is required for the calibrated result.
- A companion file, `GUIDE.md`, explains every notebook step in plain language and answers the nine comprehension questions.

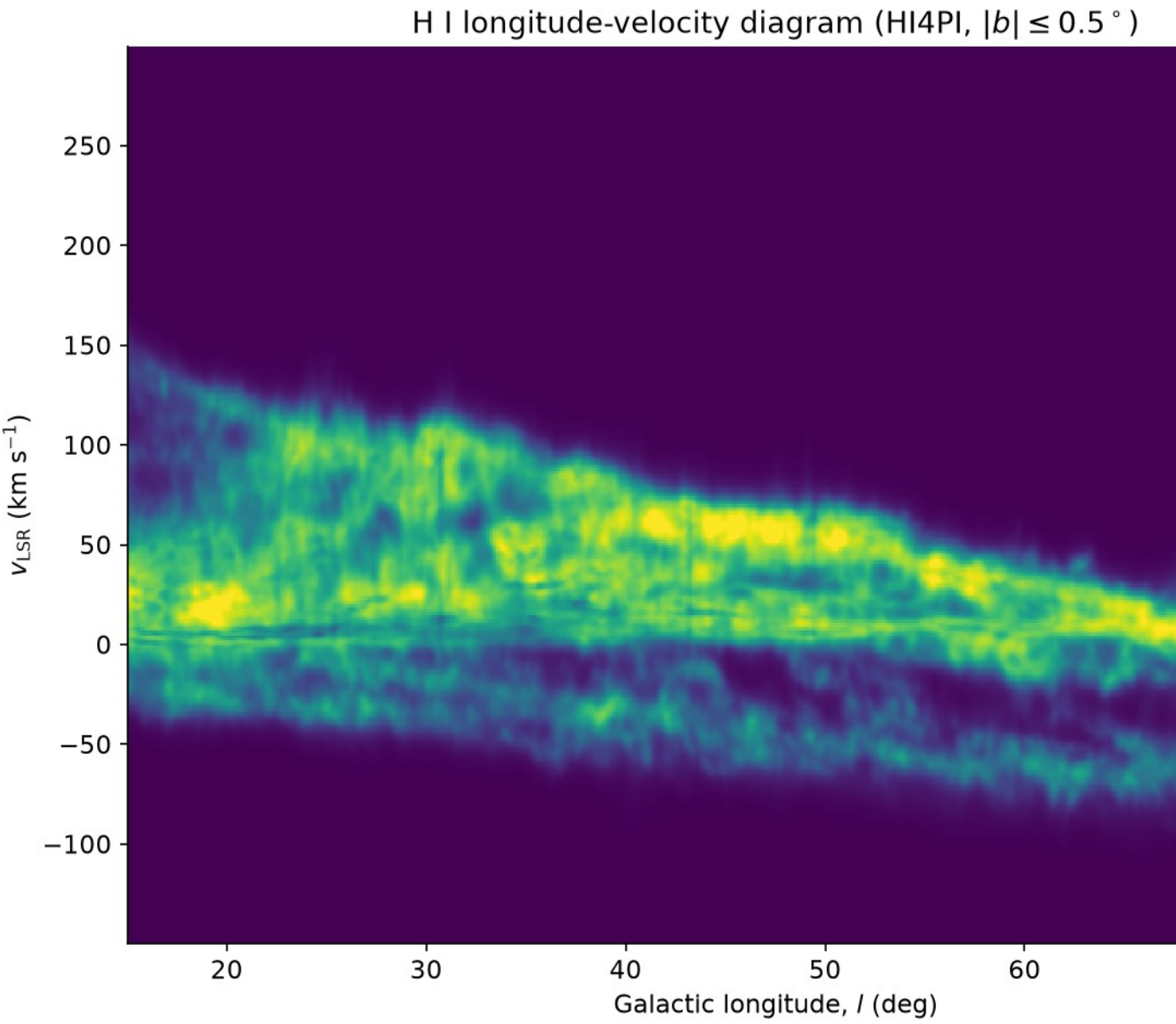
Figures

#	File	Content
1	`01_lv_diagram.png`	l–v diagram, the parent of every measurement
2	`02_sample_spectra.png`	Six real H I spectra
3	`03_terminal_velocity_overlay.png`	Quality control: measured v_term hugs the emission edge
4	`04_threshold_on_spectra.png`	Where the 10 K criterion places the edge
5	`05_rotation_curve.png`	The rotation curve, Θ(R)
6	`06_enclosed_mass.png`	Enclosed mass M(<R) vs the flat-curve expectation
7	`07_measured_vs_keplerian.png`	Measurement vs the Keplerian point-mass curve
8	`08_full_rotation_curve.pn`	Inner measurement joined to Eilers et al.

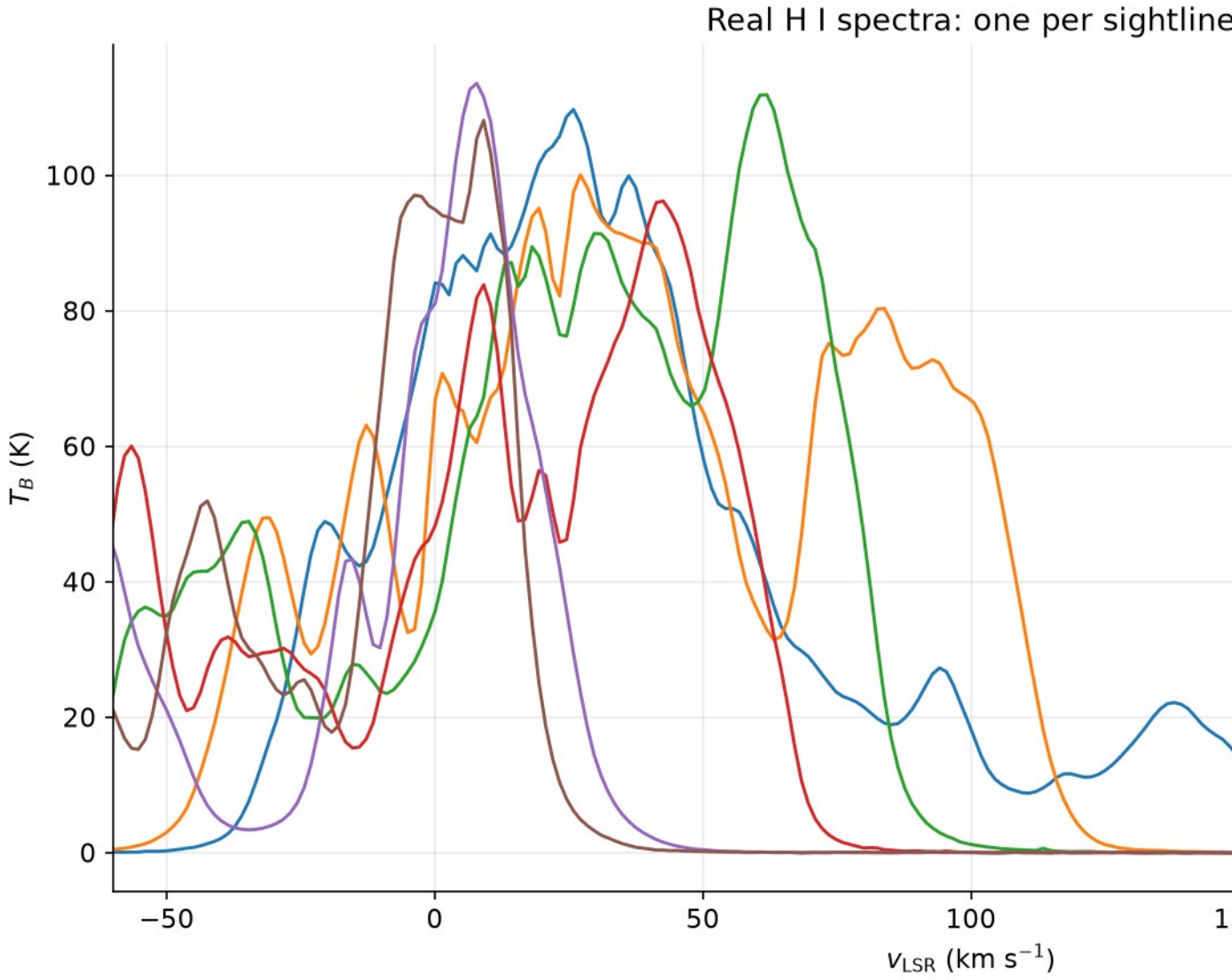
#	File	Content
	`g`	(2019)
9	`10_edge_bias_calibration.png`	The high-longitude diagnostic, l ≥ 70°
10	`11_before_and_after_correction.png`	Shape unchanged, zero point moved by 13.6 km/s

Figures

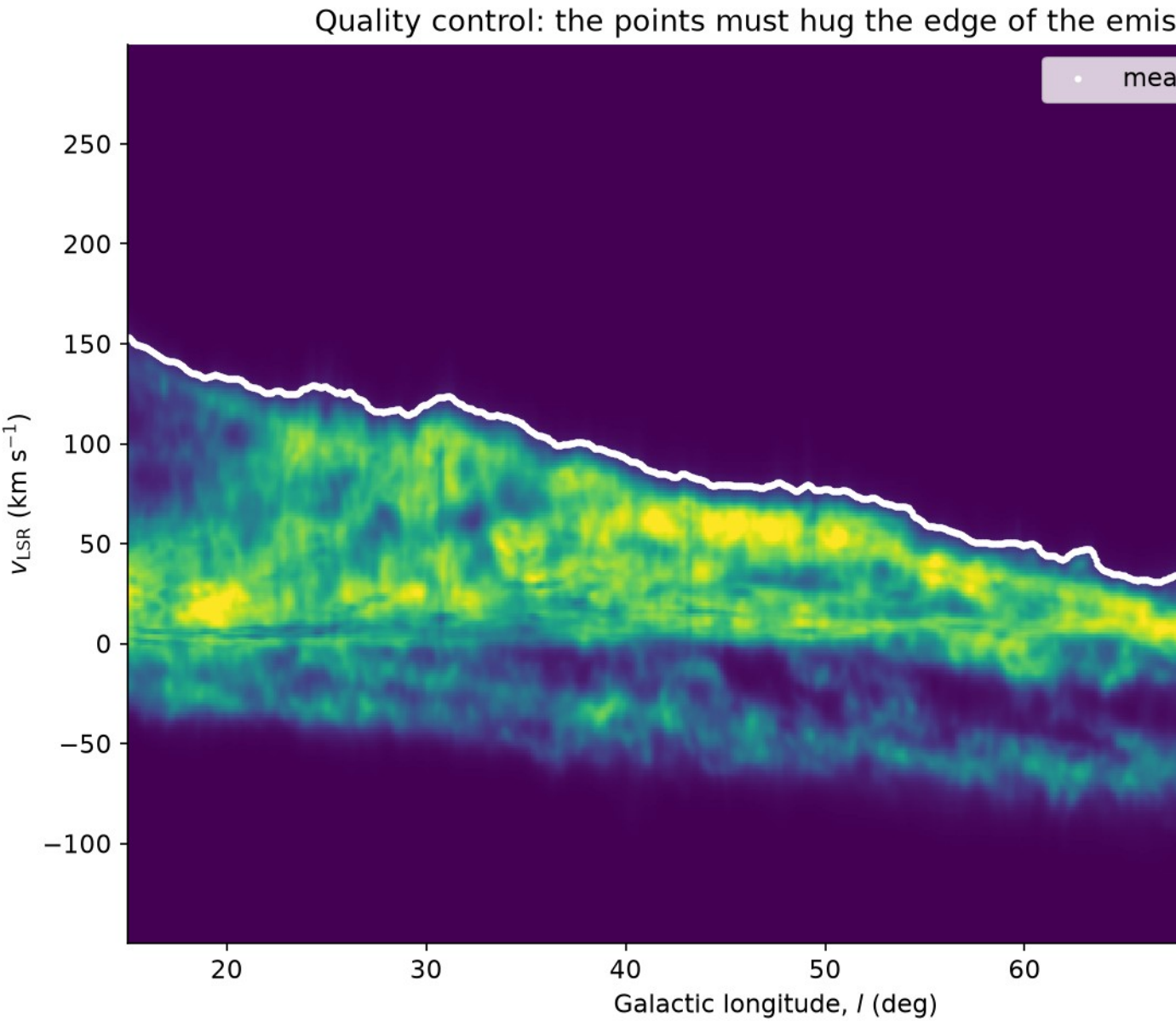
01_lv_diagram.png



02_sample_spectra.png

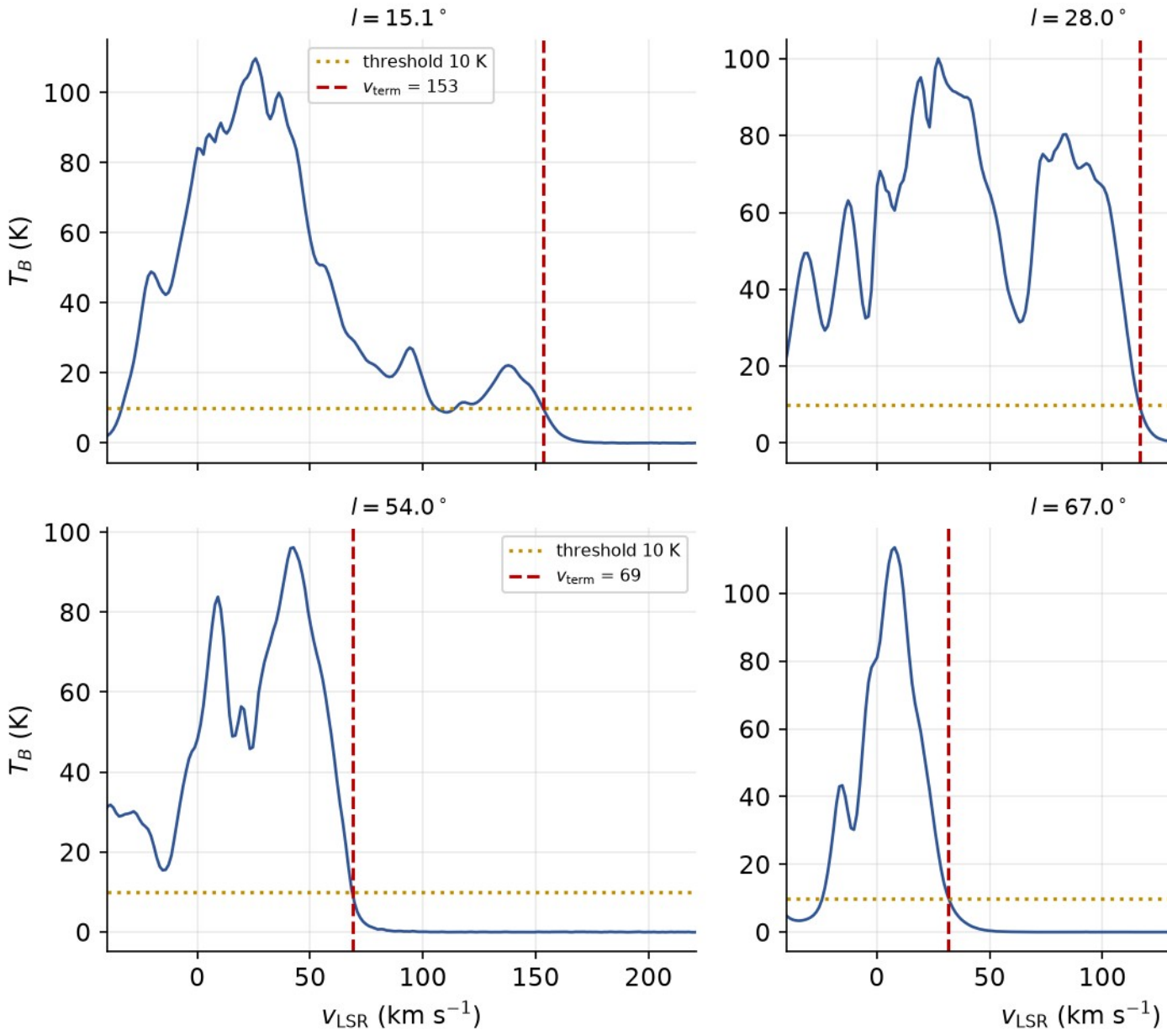


03_terminal_velocity_overlay.png

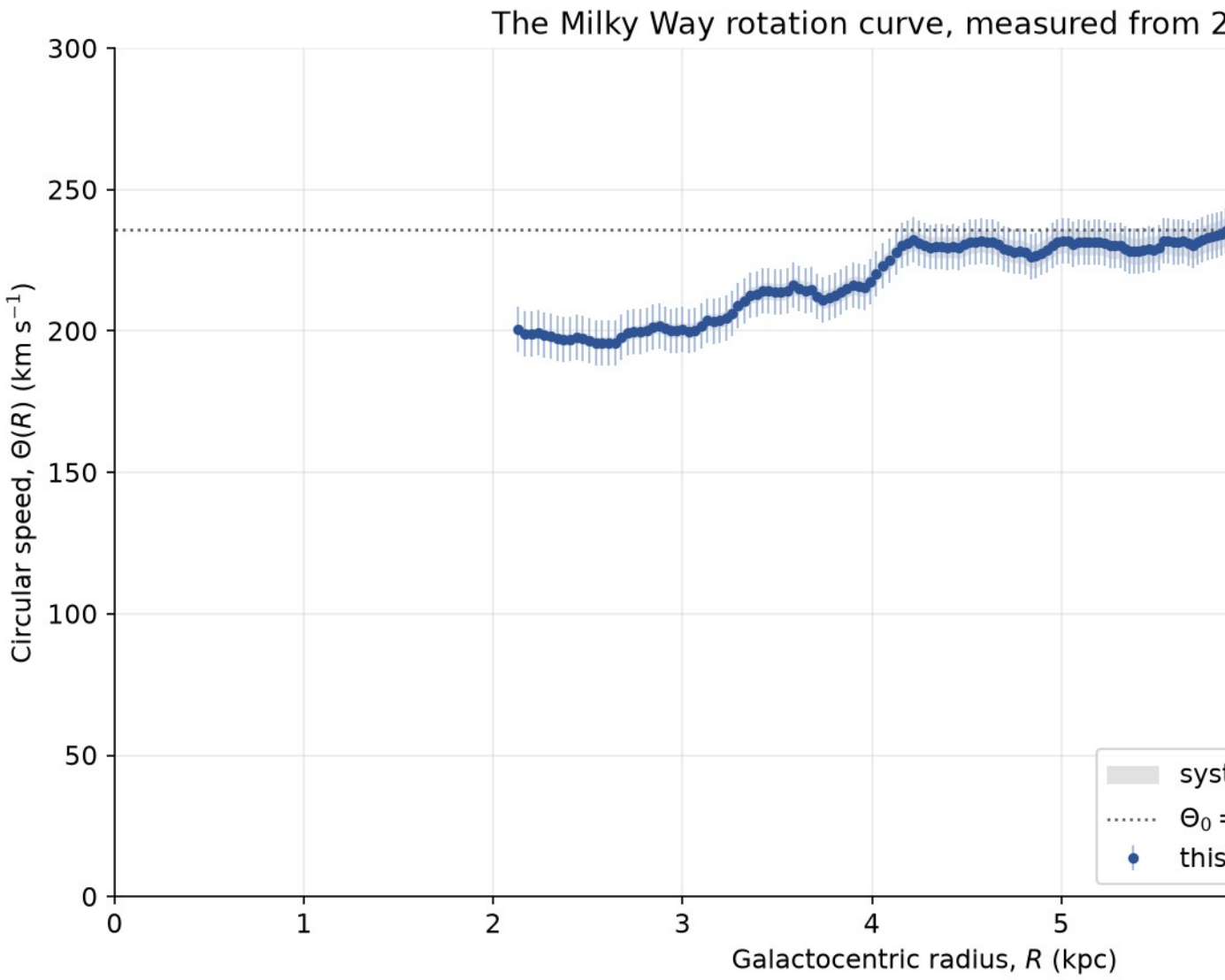


04_threshold_on_spectra.png

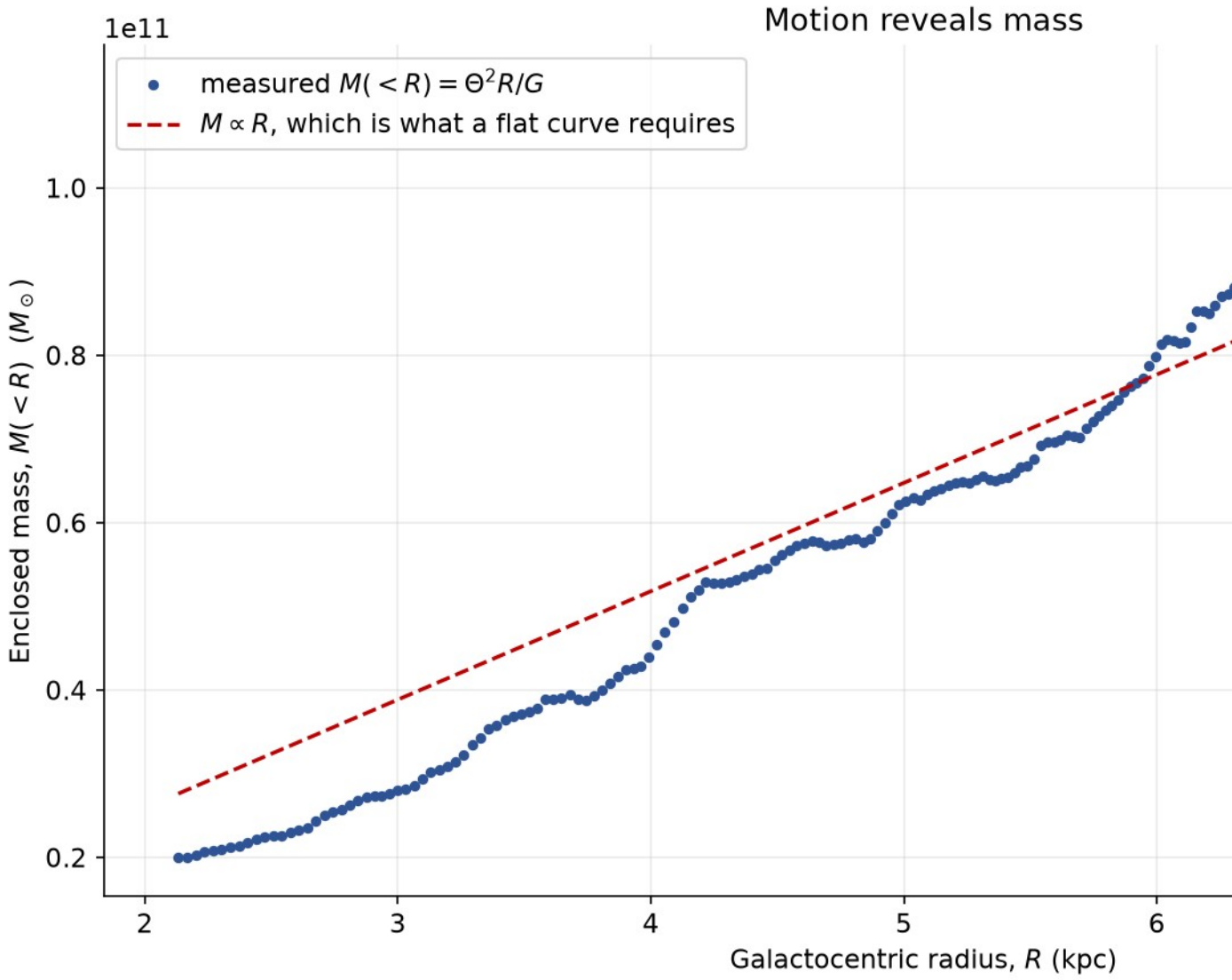
Where the criterion places



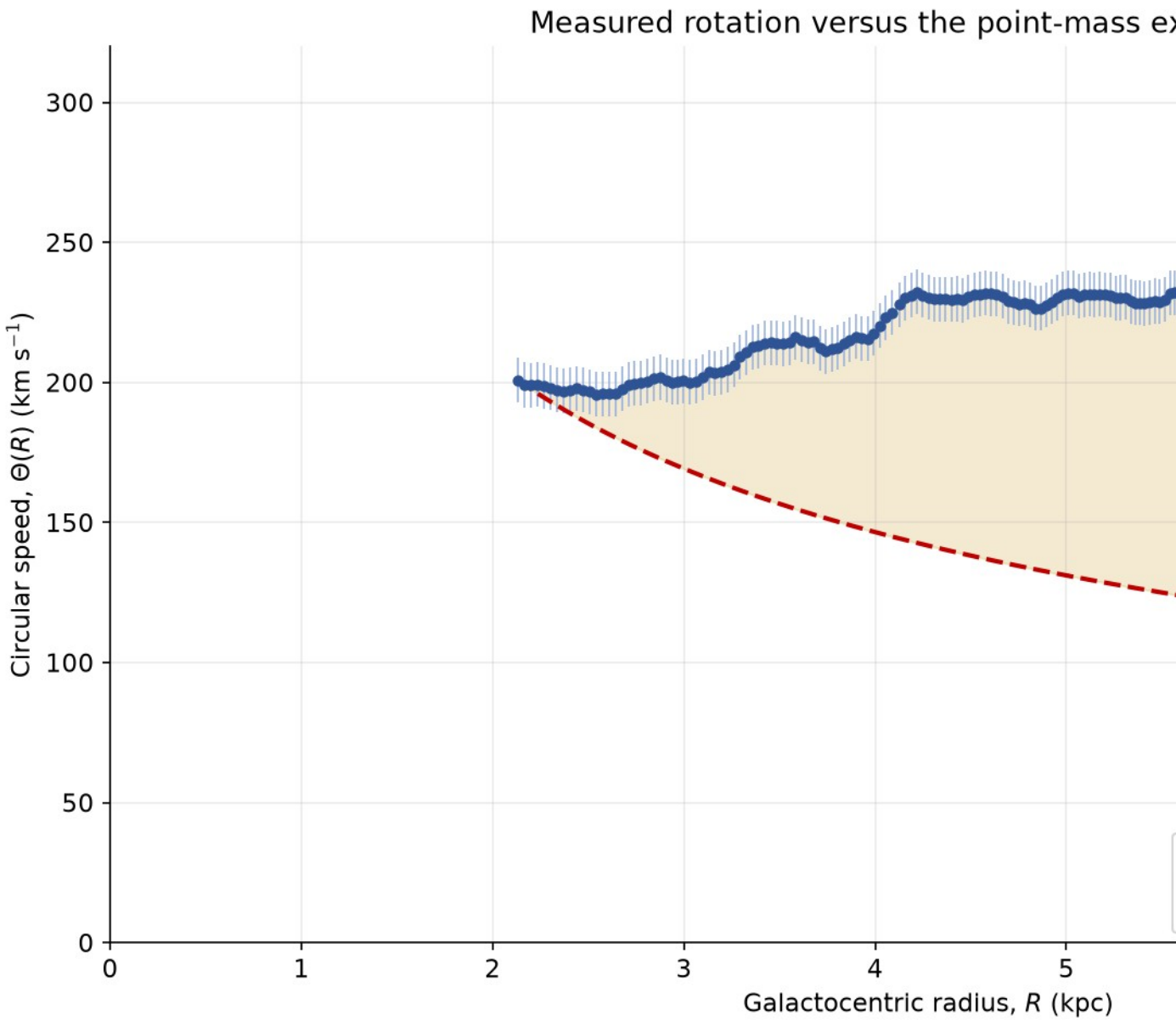
05_rotation_curve.png



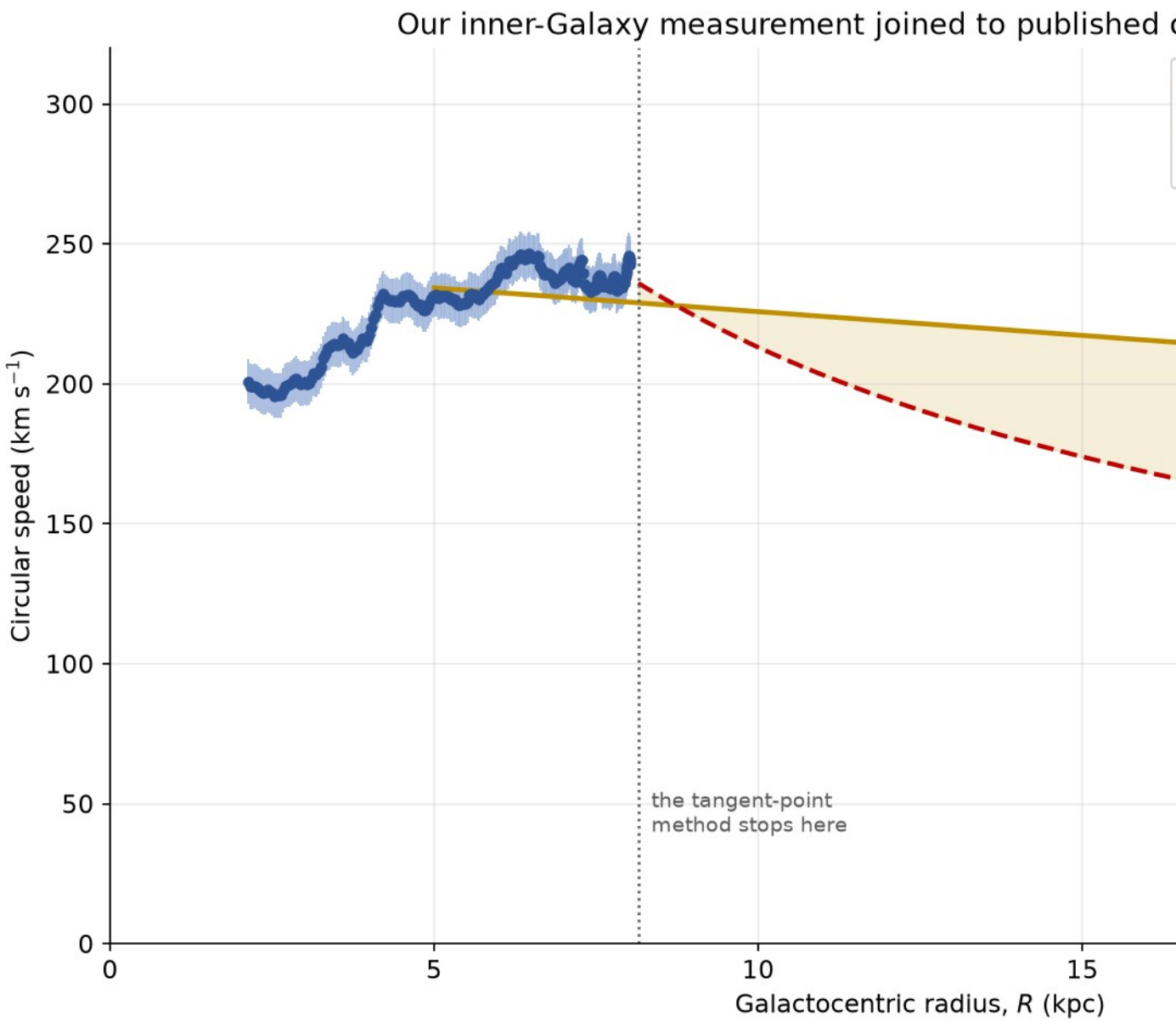
06_enclosed_mass.png



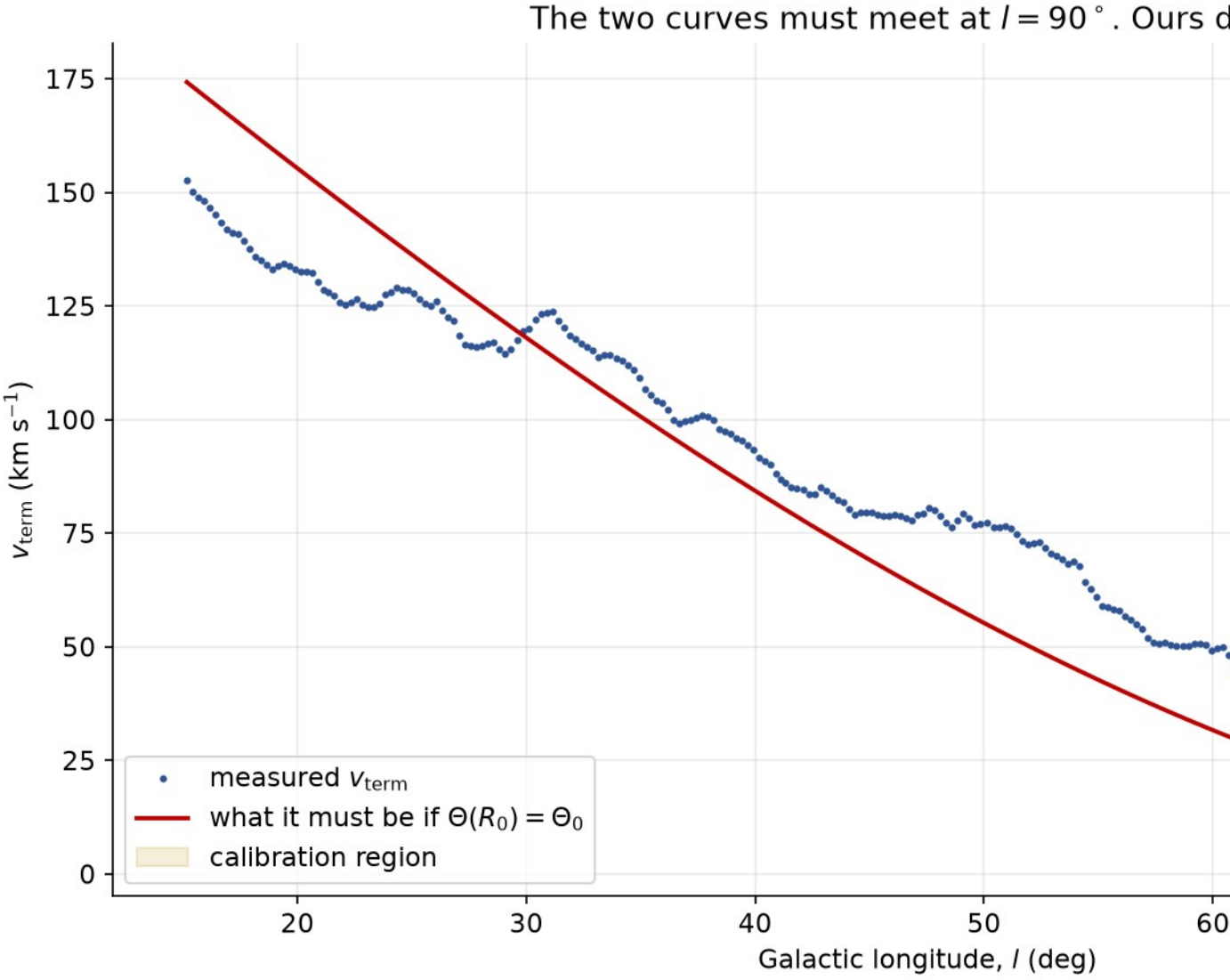
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